

Draft-then-Refine WiFi 3D Pose Estimation with Compact Mamba2 Encoding and Two-Step Flow

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Abstract—WiFi-based human pose estimation offers an occlusion-resilient alternative to camera-based sensing, but estimating multi-person 3D skeletons from channel state information remains difficult because WiFi signals are noisy, low-dimensional, and strongly affected by multipath propagation. This paper studies a compact draft-to-refine redesign of the Person-in-WiFi 3D pipeline. Starting from the PETR-style baseline, we introduce motion-aware spectral tokenization, a flattened Mamba2-CSI encoder, a lightweight WiTiDAR pose head, and two-step flow-based inference. Under the internal 20-epoch ablation protocol, the selected M9_RF2 configuration achieves 165.487 mm MPJPE, improving over the M0 baseline by 7.053 mm while increasing throughput from 118.7 FPS to 209.9 FPS and reducing parameters from 13.130M to 3.618M. The claim is deliberately bounded to this internal ablation setting; the results support an accuracy-efficiency trade-off rather than broad superiority over external systems.

Index Terms—WiFi sensing, CSI, 3D human pose estimation, Mamba2, flow matching, spectral tokenization.

I. INTRODUCTION

WiFi channel state information (CSI) provides a non-visual sensing signal that can be used for human perception in indoor environments. Unlike camera-based pose estimation, WiFi sensing can operate under occlusion and low-light conditions, but the input is indirect: body motion perturbs multipath propagation rather than producing explicit visual evidence. This makes multi-person 3D pose estimation from WiFi substantially harder than pose estimation from images.

Person-in-WiFi 3D demonstrated an end-to-end PETR-style formulation for multi-person 3D pose estimation from WiFi CSI [1]. In this work, we ask whether that pipeline can be made smaller and faster while preserving or improving pose accuracy. Our redesign follows a staged path from the original baseline to a compact draft-to-refine model. The final selected model, M9_RF2, combines spectral tokenization, Mamba2-CSI sequence modeling, a WiTiDAR draft head, and two-step flow refinement at inference.

The contribution is an internally controlled ablation rather than a new benchmark claim. We report the effect of each architectural change, separate one-step and two-step flow inference, and include supplementary analysis of missed-person behavior. The evidence indicates that two-step inference on the trained M9 checkpoint is a better operating point than the original one-step flow solver.

II. RELATED WORK

Camera-based human pose estimation has been extensively studied through bottom-up and top-down pipelines, including OpenPose-style part affinity fields [2]. Transformer architectures have also become important for structured prediction after the success of attention-based sequence modeling [3]. Person-in-WiFi 3D adapts this detection-and-regression perspective to WiFi CSI and provides the baseline used in this study [1].

State space models such as Mamba provide linear-time sequence modeling with selective recurrence [4]. Mamba-2 further connects structured state space models and attention-like computation [5]. These properties motivate replacing a heavier Transformer encoder with a compact Mamba2-CSI encoder for WiFi token sequences.

Rectified flow learns a velocity field for transporting samples between source and target distributions [6]. Here, the source is not random noise; it is a draft 3D pose produced by the detection head. The flow module is therefore used as a learned pose refinement operator. Because no reflow stage is performed, the interpretation is closer to conditional flow-matching refinement than a full generative rectified-flow pipeline.

III. METHOD

A. Overview

Figure 1 summarizes the pipeline. A preprocessed CSI tensor is converted into WiFi tokens, encoded by a compact sequence model, decoded into person hypotheses, and refined from draft pose to final pose through a flow-based correction module.

B. Motion-Aware Spectral Tokenization

The baseline uses a linear projection from CSI features to token embeddings. The spectral tokenizer keeps this linear path as a residual bypass and adds a motion-aware temporal branch. The projected sequence is reshaped into spatial groups and temporal steps. A depthwise temporal convolution smooths local temporal variation, and a real FFT over time estimates a spectral motion profile. A lightweight frequency-gating module maps this profile back to time-step weights.

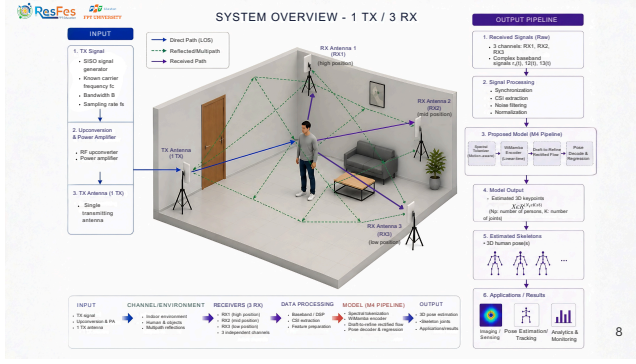


Fig. 1. System overview adapted from the project presentation deck.

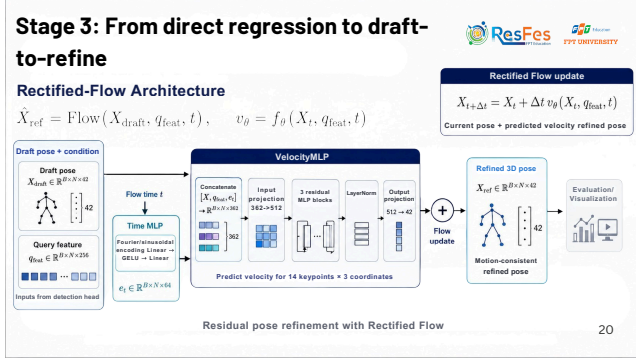


Fig. 2. Draft-to-refine flow module adapted from the project presentation deck.

C. Mamba2-CSI Encoder

The Mamba2-CSI encoder replaces a Transformer-heavy encoder with a compact sequence model. In the selected flattened configuration, tokens are processed as a single WiFi sequence rather than through multiple manually routed branches. This reduces parameter count and latency while retaining sequence context across spatial and temporal CSI tokens.

D. WiTiDAR Draft Head and Flow Refinement

The WiTiDAR head predicts person hypotheses through learned pose queries. During training, Hungarian assignment and the classification, keypoint, and bone losses are computed on draft predictions. For matched positives, the flow module receives the detached draft pose, the target pose, and the query condition. The training objective follows a conditional velocity-matching form:

$$x_t = (1 - t)x_0 + tx_1, \quad (1)$$

$$v^*(x_t, t, c) = x_1 - x_0, \quad (2)$$

$$\mathcal{L}_{\text{flow}} = \|v_\theta(x_t, t, c) - v^*(x_t, t, c)\|_2^2. \quad (3)$$

At inference, Euler integration updates the pose:

$$x_{k+1} = x_k + \Delta t v_\theta(x_k, t_k, c). \quad (4)$$

M9 uses one step, while M9_RF2 uses two steps on the same trained checkpoint.



Fig. 3. Main ablation visualization for MPJPE, latency, and parameter count.

IV. EXPERIMENTAL PROTOCOL

All models are evaluated under the same internal Person-in-WiFi 3D 20-epoch ablation protocol. The primary metric is MPJPE in millimeters. We also report one-person, two-person, and three-person splits, latency, FPS, parameter count, and memory. Runtime is measured with the project benchmark script using CUDA inference. The present manuscript does not include external SOTA comparison or repeated-seed variance.

V. RESULTS

A. Main Ablation

Table I reports the main development path. M0 is the PETR-style baseline. M1 isolates spectral tokenization. M7 replaces PETRHead with the WiTiDAR head under a Transformer encoder. M8 and M9 compare Mamba-family encoders. M9_RF2 evaluates the trained M9 checkpoint with two flow refinement steps.

M9_RF2 improves MPJPE by 7.053 mm over M0 and by 2.599 mm over the one-step M9 inference. It also reduces parameters by 72.45% relative to M0.

B. Flow Solver Analysis

Table II separates the draft-only model, no-flow inference, one-step flow, two-step flow, and retrained flow-loss variants. The two-step solver is the best observed operating point for the original M9 checkpoint, while four-step inference on the retrained branch does not improve results.

VI. DISCUSSION

The results support a narrow but useful conclusion: the selected combination of spectral tokenization, flattened Mamba2-CSI encoding, WiTiDAR draft prediction, and two-step flow inference gives the best internal accuracy-efficiency trade-off among the tested configurations. The results do not imply that more flow steps always improve performance. In fact, the non-monotonic behavior from one to two to four steps

TABLE I
MAIN 20-EPOCH ABLATION FROM PETR-STYLE BASELINE TO SELECTED COMPACT MODEL. LOWER MPJPE IS BETTER.

ID	Role	Input	Encoder	Head	Flow	MPJPE	1P	2P	3P	FPS	Params
M0	Baseline	Linear	Transformer	PETRHead	No	172.540	126.059	168.183	196.492	118.7	13.130M
M1	Spectral baseline	Spectral	Transformer	PETRHead	No	169.271	125.502	165.038	193.273	83.1	13.199M
M7	WiTiDAR Transformer	Spectral	Transformer	WiTiDAR	1-step	168.342	127.129	162.775	191.390	145.8	7.059M
M8	WiTiDAR Mamba-1	Spectral	Mamba-1	WiTiDAR	1-step	169.653	122.253	165.397	193.943	109.4	5.827M
M9	WiTiDAR Mamba2	Spectral	Mamba2-CSI	WiTiDAR	1-step	168.086	125.958	161.832	192.229	206.9	3.618M
M9_RF2	Selected	Spectral	Mamba2-CSI	WiTiDAR	2-step	165.487	121.785	159.341	190.179	209.9	3.618M

TABLE II
SUPPLEMENTARY FLOW ABLATION. LOWER MPJPE AND MATCHED-ONLY MPJPE ARE BETTER.

ID	Setting	MPJPE	Matched MPJPE	Missed	1P	2P	3P
M6	Draft Mamba2-CSI	167.573	166.381	54	125.532	164.528	188.363
M9_no_flow	M9 checkpoint, flow disabled	167.732	165.919	82	122.942	161.484	192.178
M9	One-step flow	168.086	166.409	76	125.958	161.832	192.229
M9_RF2	Two-step flow	165.487	163.886	72	121.785	159.341	190.179
T_FW2_20e_RF2	Flow loss weight 2.0, two-step	167.981	166.082	86	121.335	160.970	194.804
T_FW2_20e_RF4	Flow loss weight 2.0, four-step	169.454	167.275	99	122.157	162.857	196.141

deployment claims require cross-room, cross-subject, and hardware-aware evaluation.

VIII. CONCLUSION

This paper presents a compact draft-to-refine WiFi 3D pose pipeline. Under the internal ablation protocol, M9_RF2 achieves 165.487 mm MPJPE, improves over the M0 PETR-style baseline by 7.053 mm, reaches 209.9 FPS, and reduces parameters to 3.618M. These results indicate that compact sequence modeling and carefully selected flow inference can improve WiFi-based multi-person 3D pose estimation without increasing model complexity.

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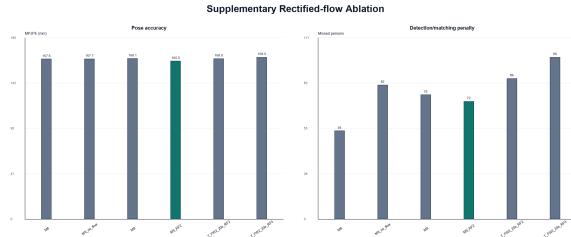


Fig. 4. Flow solver ablation showing MPJPE and missed-person count.

suggests that the learned velocity field is not perfectly straight and that solver choice must be validated carefully.

The comparison with M6 shows a detection-refinement trade-off. M6 has fewer missed persons but worse overall MPJPE than M9_RF2. This motivates future work on refinement-aware confidence calibration, because inference still ranks candidates using draft-stage confidence while reporting refined coordinates.

VII. LIMITATIONS

This study is limited to an internal 20-epoch ablation protocol. It does not report repeated random seeds, validation-based solver selection, or external leaderboard comparison. The term flow refinement is used in a conditional correction sense; no reflow stage is performed. The spectral branch is best described as motion-aware rather than a direct Doppler estimator because it operates after feature projection. Broader